

NELSR: A Nepali Legal Benchmark for Case-to-Statute Retrieval

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Abstract

Legal professionals spend considerable time identifying statutes relevant to legal cases by navigating lengthy and hierarchically structured legal documents. While automated statute retrieval has advanced for high-resource languages to resolve the problem, the Nepali legal domain still lacks a standardized benchmark for realistic case-to-statute retrieval and systematic evaluation. To address this gap, we introduce NELSR (Nepali Legal Statute Retrieval Benchmark), comprising 5,731 case-derived queries extracted from Supreme Court judgments and a corpus of 302 federal statutes. NELSR is constructed using a citation-aware query construction approach that establishes reliable query–statute relevance pairs and legal structure-aware segmentation to preserve the semantic and hierarchical organization of statutory units. We establish retrieval baselines by evaluating sparse, dense, and hybrid retrieval methods and investigate the contributions of lexical and semantic retrieval signals through ablation studies. Experimental results show that hybrid retrieval achieves the best performance, with a Recall@100 of 0.66, nDCG@10 of 0.34, and MRR@10 of 0.32.

1 Introduction

The rapid growth of digital information has created an increasing need for efficient methods to organize, search, and retrieve relevant information from large-scale document collections. Information Retrieval (IR) addresses this challenge by enabling the identification of information resources that satisfy a user’s information need from large corpora (Manning et al., 2008). IR has been widely applied across diverse domains, including healthcare, scientific literature retrieval, and legal information systems (Sivarajkumar et al., 2024; Manconi et al., 2012; Sansone and Sperlí, 2022). Among these, the legal domain represents a particularly challenging application area, where legal professionals must

efficiently locate relevant information to support legal research and decision-making. The importance of this application is further underscored by the fact that delays in legal processes can negatively impact access to justice and impose significant burdens on individuals involved in legal proceedings (Melcarne et al., 2021).

Legal Information Retrieval (Legal IR) is a specialized area of IR that focuses on retrieving relevant legal materials, such as statutes, regulations, and judicial decisions, from large legal corpora given a user query (Sansone and Sperlí, 2022). Compared to general-domain retrieval, Legal IR presents unique challenges, including complex document structures, domain-specific terminology, long document lengths, and the need for high-precision retrieval in legally meaningful contexts (Locke and Zucco, 2022). Early retrieval models, including TF-IDF (Salton and Buckley, 1988) and BM25 (Robertson and Zaragoza, 2009), addressed efficient document ranking through lexical matching and probabilistic relevance estimation. To overcome the limitations of lexical matching, neural retrieval methods leveraged dense representations and transformer-based models to capture semantic relationships between queries and documents (Reimers and Gurevych, 2019; Karpukhin et al., 2020). Domain-specific language models, such as Legal-BERT, further improved retrieval effectiveness by adapting pretrained transformers to legal corpora and terminology (Chalkidis et al., 2020).

Despite these advances, existing Legal IR benchmarks, including prominent resources such as LexGLUE (Chalkidis et al., 2022) and COLIEE (Rabelo et al., 2021), have primarily focused on legal systems and languages with relatively rich computational and linguistic resources. As a result, low-resource legal domains remain underrepresented in benchmark development and systematic evaluation. These domains face several chal-

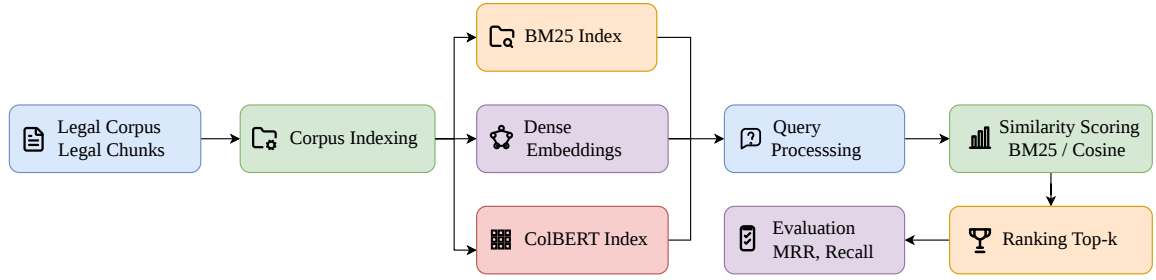


Figure 1: Retrieval pipeline.

lenges, including limited annotated data, insufficient domain-specific language resources, and substantial variations in legal terminology and document structures across jurisdictions (Krasadakis et al., 2024; Hedderich et al., 2020). Such constraints make it challenging to directly transfer existing Legal IR approaches to underrepresented legal systems. Nepali represents one such low-resource legal language. Although statutes, regulations, and judicial decisions are increasingly available in digital formats, standardized benchmarks for evaluating legal retrieval systems are still lacking. To fill this gap, we present NELSR (Nepali Legal Statute Retrieval Benchmark), comprising 5,731 case-derived queries and a corpus of 302 federal statutes for systematic evaluation of legal information retrieval in the Nepali domain. The overall experimental pipeline is illustrated in Figure 1, providing a comprehensive overview of the key stages involved in the retrieval process.

Our contributions are threefold: (1) we introduce, to the best of our knowledge, the first benchmark for Nepali case-to-statute retrieval, addressing the lack of standardized evaluation in this domain; (2) we provide a systematic comparison of sparse, dense, and hybrid retrieval methods, evaluating both effectiveness and computational efficiency; and (3) we analyze the interplay between lexical and semantic retrieval signals through detailed empirical evaluation.

2 Related Work

2.1 Retrieval Methodologies

Legal IR has evolved through several retrieval paradigms to improve access to complex legal documents. Early approaches relied on sparse lexical retrieval methods, where documents were ranked based on term matching (Manning et al., 2008). Statistical methods such as TF-IDF (Salton and Buckley, 1988) and BM25 (Robertson and Zaragoza,

2009) strengthened these approaches through term weighting and probabilistic relevance estimation, establishing strong baselines for large-scale retrieval. However, their reliance on lexical overlap limits their ability to retrieve relevant documents when queries and documents use different terminology. To mitigate this limitation, dense retrieval methods represent queries and documents as continuous vector embeddings, enabling semantic matching beyond exact term overlap (Reimers and Gurevych, 2019; Karpukhin et al., 2020). Building on these advances, domain-specific transformer models such as Legal-BERT (Chalkidis et al., 2020) further improve retrieval by adapting contextual representations to legal terminology and domain-specific language patterns. Despite these improvements, single-vector dense representations may overlook fine-grained token-level interactions. To address this, late-interaction models such as ColBERT (Khattab and Zaharia, 2020) preserve token-level matching while maintaining efficient retrieval. Recognizing that lexical and semantic signals offer complementary strengths, hybrid retrieval approaches combine sparse and dense methods to achieve more robust performance (Luan et al., 2021). More recently, graph-based retrieval methods have incorporated structural signals from legal corpora, such as citation networks and inter-document relationships, enabling richer representations that further enhance retrieval effectiveness (Chen et al., 2026).

2.2 Legal IR Benchmarks

The development of benchmark datasets has been critical for advancing Legal IR research. The COLIEE benchmark (Rabelo et al., 2021) is one of the most established evaluation platforms, featuring tasks such as case law retrieval and statute retrieval. Large-scale datasets such as LeCaRD (Ma et al., 2021) and LeCaRDv2 (Li et al., 2024) have further advanced research by providing manually anno-

tated case retrieval datasets in the Chinese legal domain. Similarly, the CLERC (Case Law Evaluation and Retrieval Corpus) benchmark (Hou et al., 2025) supports evaluation of U.S. legal case retrieval and retrieval-augmented reasoning tasks. In the legislative domain, resources such as EUR-Lex (Aumiller et al., 2022) and MultiEURLEX (Chalkidis et al., 2021) provide large-scale collections of European Union legal documents, enabling research in multilingual legal NLP.

Despite these advances, existing Legal IR benchmarks are predominantly focused on high-resource languages and well-established legal systems. Low-resource legal domains remain significantly underrepresented, with limited availability of benchmark datasets for statute-level retrieval tasks. This gap motivates the development of benchmarks tailored to underrepresented legal systems, such as Nepali Legal IR.

3 Dataset

3.1 Data Collection

To construct the statutory corpus, we collected 302 federal statutes from the official website of the Ministry of Law, Justice, and Parliamentary Affairs of Nepal (Ministry of Law, Justice and Parliamentary Affairs, 2026). The collection includes 301 acts and the Constitution of Nepal, provided in DOCX and DOC formats. For query construction, we collected 10,464 judicial decisions from Nepal Kanoon Patrika (NKP) (Nepal Kanoon Patrika, 2026), a legal information portal containing Supreme Court judgments. These judgments provide realistic legal contexts from which citation-based queries and query-statute relevance pairs are derived.

3.2 Chunk Construction

The collected federal legal documents were processed using a heuristic-based parsing pipeline to transform unstructured legal texts into structured retrieval units. The documents were first extracted into paragraphs and subjected to text normalization, including Unicode normalization, removal of formatting artifacts introduced during document conversion, and whitespace standardization. Regular expression-based rules were then applied to identify the hierarchical structure of legal documents, including articles, sub-articles, and clauses.

Articles containing fewer than 200 words were retained as complete article-level chunks. This threshold was selected based on an analysis of ar-

ticle length distribution and legal document structure, where shorter articles generally represent self-contained legal provisions, while longer articles often contain multiple sub-articles or clauses requiring finer segmentation. For longer articles, the pipeline prioritized existing legal structures by splitting them at sub-article or clause boundaries whenever such structures were available. When no explicit structural boundaries were detected, articles were divided into balanced word-based chunks.

3.3 Query Construction

To construct retrieval queries, we utilized judicial decisions collected from the NKP website. The case documents were processed to extract the main judgment body by identifying decision sections and excluding preliminary metadata and post-judgment administrative content.

Legal citations within the judgment text were automatically detected using a rule-based citation extraction method that identifies referenced acts and their corresponding sections. For each detected citation, the sentence containing the citation along with its surrounding sentence context was extracted to form the query text. The citation itself was removed from the query text to prevent direct lexical matching between the query and the target legal provision. To improve benchmark quality, we remove duplicate queries using TF-IDF (Salton and Buckley, 1988) similarity and filter outliers in citation-context lengths using the standard interquartile range (IQR) method. Finally, the queries were linked with corresponding article and sub-article identifiers to generate relevance judgments for benchmark evaluation.

3.4 Dataset Statistics

The overall statistics of the constructed benchmark are summarized in Table 1. The legal corpus consists of 302 legal documents transformed into 22,326 retrieval chunks. The query set contains 5,731 citation-derived queries extracted from 10,464 judicial decisions.

The relevance annotations provide the mapping between queries and their corresponding legal provisions, with statistics reported based on the number of relevant chunks associated with each query. The benchmark exhibits a sparse relevance distribution, where most queries are associated with a single relevant chunk, while only a small number of queries contain multiple relevant chunks. The distribution of relevant chunks per query is

Statistic	Value
Legal documents	302
Retrieval chunks	22,326
Median chunk length (words)	41
Min / Max chunk length (words)	6 / 256
Judicial decisions	10,464
Queries	5,731
Median query length (words)	128
Min / Max query length (words)	17 / 577
Avg. relevant chunk per query	1
Min / Max target chunks	1 / 4

Table 1: Statistics of the NELSR dataset. Query statistics are based on sentence spans comprising the citation-containing sentence, one sentence before, and one sentence after.

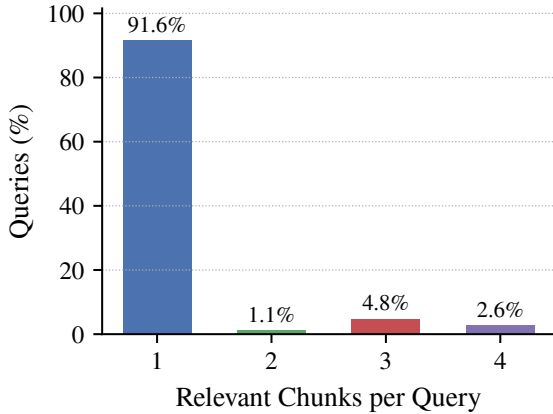


Figure 2: Percentage distribution of target spans per query

illustrated in Figure 2, highlighting the characteristics of the proposed benchmark for retrieval evaluation. The released benchmark resources, including processed queries, statute chunks, and experimental code, are available in an anonymous repository for review: <https://anonymous.4open.science/r/NELSR-EEDE/>.

4 Retrieval Framework

Given a legal query, the goal of the retrieval system is to rank relevant statutory chunks from the corpus. All evaluated approaches operate on the same corpus, queries, and relevance judgments, differing only in their scoring mechanisms.

Sparse retrieval uses BM25 (Robertson and Zaragoza, 2009), implemented with an inverted index for efficient term-based matching. Dense retrieval methods encode queries and legal chunks into vector representations and perform similarity

search using FAISS (Johnson et al., 2017). Late-interaction retrieval (ColBERT) preserves token-level representations and employs its native indexing along with a MaxSim-based matching mechanism (Khattab and Zaharia, 2020). Hybrid retrieval combines sparse and dense scores through weighted fusion, leveraging FAISS for dense components and BM25 for lexical matching.

For all approaches, retrieved chunks are ranked by their relevance scores, and the top- k results are evaluated using standard retrieval metrics. In addition to effectiveness, we analyze retrieval efficiency in terms of indexing time, retrieval latency, and index size. The retrieval pipeline is illustrated in Figure 1.

5 Experiments

In this section, we evaluate the proposed benchmark using existing retrieval frameworks and address the following research questions:

RQ1: How effective are multilingual and Nepali-specific dense retrieval models for case-to-statute retrieval in the Nepali legal domain?

RQ2: How do sparse, dense, late-interaction, and hybrid retrieval approaches compare for retrieving relevant statutory provisions from Nepali judicial decisions?

RQ3: What are the computational trade-offs among different retrieval approaches in terms of index size, retrieval efficiency, and indexing time for Nepali case-to-statute retrieval?

RQ4: How can lexical and semantic signals be effectively balanced to improve statutory provision retrieval from Nepali legal cases?

5.1 Experimental Setup

Metrics and Data. We evaluate retrieval performance on the proposed Nepali Legal IR benchmark using the data described in 3. Retrieval systems are assessed using standard information retrieval metrics, including Mean Reciprocal Rank (MRR), Recall@ k , and Normalized Discounted Cumulative Gain (nDCG@ k). All methods are evaluated on the same corpus, queries, and relevance judgments to ensure a fair comparison.

Implementation Details. All experiments were conducted using Google Colab free-tier computational resources with approximately 15 GB of available memory and access to a GPU.

5.2 Query Span Selection

We define the query span as the sentence-level context surrounding a cited legal reference. The span includes the sentence containing the citation along with a configurable number of preceding and following sentences. To determine an effective query span, we perform a query span ablation study using BM25 (Robertson and Zaragoza, 2009) retrieval. We evaluate multiple sentence-span configurations, ranging from the citation-containing sentence alone to configurations including up to three sentences before and after the citation. Each configuration is evaluated on the same query set using Recall@5, Recall@10, and Recall@100. The retrieval results are shown in Table 2

The results show that incorporating surrounding sentences generally improves retrieval effectiveness compared with using only the citation-containing sentence. Including one sentence before and one sentence after the citation achieves the best early retrieval performance, obtaining the highest Recall@5 (0.3591) and Recall@10 (0.4172). However, expanding the context further provides limited gains at lower ranks, likely because additional sentences introduce less relevant legal information. The highest Recall@100 is achieved by including one sentence before and two sentences after the citation (0.6088), but the improvement over the selected configuration is marginal. Therefore, we select the one-before and one-after sentence span as the default query construction strategy, providing a balance between retrieval effectiveness and contextual compactness.

5.3 RQ1: Multilingual vs. Nepali-Specific Dense Retrieval Models

To investigate the effectiveness of multilingual and Nepali-specific dense retrieval models on NELSR, we evaluate six embedding-based retrievers. The evaluated models include Nepali-specific models (UniversalML-Nepali (Singh, 2025), Nepali-Embedder-v1 (Pathak, 2026), and SemantiSearch-NepaliSBERT (Pachhain, 2025)) and multilingual models (BGE-M3 (Chen et al., 2024), multilingual-E5-Large (Wang et al., 2024), and Jina Embeddings (Gunther et al., 2024)). The results are presented in Table 3.

Within dense retrievers, multilingual models generally surpass Nepali-specific ones. BGE-M3 delivers the strongest overall dense retrieval results, achieving the highest MRR@10 (0.21) and

nDCG@10 (0.23), while Jina Embeddings attains the top Recall@10 (0.33) and Recall@100 (0.54). Among models tailored to Nepali, SemantiSearch-NepaliSBERT performs best, with a Recall@100 of 0.55 and competitive effectiveness at lower retrieval depths.

Overall, the results indicate that multilingual models provide stronger semantic representations on NELSR, while Nepali-specific models remain competitive, particularly at deeper retrieval ranks. Since hybrid retrieval aims to improve candidate retrieval coverage by combining lexical and semantic signals, we select Jina Embeddings and SemantiSearchNepaliSBERT as representative multilingual and Nepali-specific dense retrievers based on their stronger recall performance.

5.4 RQ2: Sparse, Dense, Late-Interaction, and Hybrid Retrieval

To compare different retrieval paradigms on NELSR, we evaluate sparse retrieval, dense retrieval models, a late-interaction model, and hybrid approaches that combine lexical and semantic signals. As shown in Table 3, BM25 establishes a strong baseline, achieving Recall@5 of 0.36, Recall@10 of 0.42, MRR@10 of 0.30, and nDCG@10 of 0.32. These results suggest that lexical matching remains effective on NELSR, likely because the benchmark contains explicit legal terminology and statutory references that align well with sparse retrieval. Dense retrievers provide complementary semantic matching capabilities, with Jina Embeddings and BGE-M3 achieving the strongest dense retrieval performance. In contrast, the late-interaction model Jina-ColBERT-v2 (Jha et al., 2024) achieves lower effectiveness, which may be attributed to the absence of domain-specific training data for Nepali legal text and the resulting challenges in learning fine-grained token-level interactions. Hybrid retrieval achieves the strongest overall performance by integrating lexical and semantic signals. The BM25 + Jina Embeddings combination provides the best early retrieval effectiveness, achieving Recall@5 of 0.39 and Recall@10 of 0.45, while BM25 + SemantiSearch-NepaliSBERT obtains the highest Recall@100 of 0.66. These findings demonstrate that sparse and dense retrieval approaches capture complementary relevance signals, and combining them improves retrieval effectiveness on NELSR.

Query Context	R@5	R@10	R@100
Cited sentence	0.3458	0.3951	0.5550
+1 sentence before	0.3488	0.4051	0.5822
+1 sentence after	0.3545	0.4110	0.5877
+1 before +1 after	0.3591	0.4172	0.6008
+2 before +1 after	0.3563	0.4152	0.6006
+1 before +2 after	0.3545	0.4114	0.6088
+2 before +2 after	0.3542	0.4075	0.6033
+3 before +2 after	0.3495	0.4095	0.6040
+2 before +3 after	0.3470	0.4043	0.5999
+3 before +3 after	0.3448	0.4019	0.6020

Table 2: BM25 retrieval performance under different sentence-based query span configurations.

Model	R@5	R@10	R@100	MRR@10	nDCG@10
Sparse Retrieval					
BM25	0.36	0.42	0.60	0.30	0.32
Dense Retrieval (Nepali)					
UniversalML-Nepali	0.26	0.31	0.47	0.10	0.21
Premmm/Nepali-Embedder-v1	0.10	0.13	0.27	0.07	0.08
SemanticSearchNepaliSBERT	0.24	0.31	0.55	0.18	0.20
Dense Retrieval (Multilingual)					
BGE-M3	0.26	0.32	0.52	0.21	0.23
Jina Embeddings	0.26	0.33	0.54	0.20	0.23
E5-Large	0.24	0.28	0.46	0.19	0.20
Late Interaction					
Jina-ColBERT-v2	0.23	0.27	0.41	0.18	0.20
Hybrid Retrieval					
BM25 + Jina ($\alpha = 0.6$)	0.39	0.45	0.64	0.32	0.34
BM25 + NepaliSBERT ($\alpha = 0.6$)	0.38	0.44	0.66	0.32	0.34

Table 3: Retrieval performance of sparse, dense, late-interaction, and hybrid retrieval approaches on NELSR. For hybrid retrieval, α represents the weight assigned to the BM25 score during score fusion. Best results are highlighted in bold.

5.5 RQ3: Computational Trade-offs

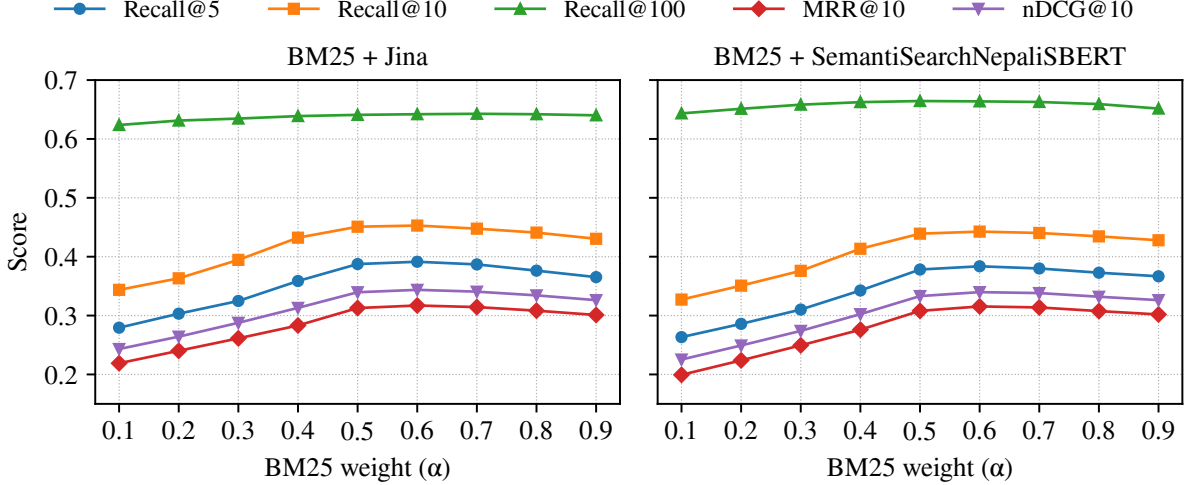
To analyze the computational requirements of dense and late-interaction retrieval models, we compare indexing time, retrieval throughput, and index size. Indexing time includes embedding generation and index construction, while index size represents the storage requirement of the retrieval index. The results are presented in Table 4. The results show clear differences in computational efficiency across retrieval models. Among dense retrievers, Nepali-Embedder-v1 is the most efficient, requiring only 147.17 seconds for indexing, achieving 39 queries/s throughput, and using 65.58 MB of storage. SemantiSearchNepaliSBERT provides a similar efficiency profile with slightly lower throughput. Multilingual models require higher computational resources, with BGE-

M3, multilingual-E5-Large, and Jina Embeddings using 87.38 MB of storage and longer indexing times.

Jina-ColBERT-v2 has the highest computational cost, requiring 1661.60 seconds for indexing, achieving only 6 queries/s throughput, and requiring 324.96 MB of storage due to token-level representations. Overall, the results demonstrate a trade-off between retrieval effectiveness and computational efficiency, where lightweight dense models provide better resource efficiency, while larger multilingual and late-interaction models incur higher computational costs. The index size remains consistent across models with identical embedding dimensionality, as embeddings are stored in 32-bit floating-point format in a flat FAISS index for dense retrievers. Sparse retrieval efficiency is ex-

Table 4: Efficiency comparison of dense and late-interaction retrieval models.

Model	Index Time (s)	Throughput (queries/s)	Index Size (MB)
Nepali-Embedder-v1	147.17	39	65.58
SemantiSearchNepaliSBERT	180.38	28	65.58
UniversalML-Nepali	626.39	13	87.38
BGE-M3	637.00	12	87.38
Jina Embeddings	972.00	11	87.38
Multilingual-E5-Large	642.00	11	87.38
Jina-ColBERT-v2	1661.60	6	324.96

Figure 3: Effect of BM25 weight (α) in hybrid score fusion with Jina Embeddings and SemantiSearchNepaliSBERT.

cluded because BM25 implementations are highly optimized and have different indexing mechanisms compared with embedding-based retrieval systems.

5.6 RQ4: Hybrid Retrieval Score Fusion Analysis

To investigate how lexical and semantic retrieval signals complement each other, we evaluate hybrid retrieval by combining BM25 with the strongest dense retrievers from multilingual and Nepali-specific categories: Jina Embeddings and SemantiSearchNepaliSBERT. We combine retrieval scores using weighted score fusion, where α controls the contribution of BM25, and evaluate fusion weights from 0.1 to 0.9 using Recall@ k , MRR@10, and nDCG@10. The effect of varying the fusion weight is shown in Figure 3. The results show that hybrid retrieval consistently improves over individual retrieval approaches by combining lexical and semantic signals. For BM25+Jina, performance improves with increasing BM25 contribution and achieves the best overall ranking performance at $\alpha = 0.6$, obtaining Re-

call@5 of 0.39, Recall@10 of 0.45, MRR@10 of 0.32, and nDCG@10 of 0.34. Similarly, BM25+SemantiSearchNepaliSBERT achieves its strongest ranking performance around $\alpha = 0.6$, while the highest Recall@100 is obtained at $\alpha = 0.5$ (0.66). These results demonstrate that lexical and semantic signals provide complementary information for Nepali legal statute retrieval, as evaluated on NELSR. BM25 effectively captures exact legal terminology, article references, and citation patterns, while dense retrievers improve semantic matching beyond lexical overlap. The optimal performance is achieved through balanced score fusion, where approximately 60% weight is assigned to lexical retrieval and 40% to semantic retrieval.

6 Conclusion

We introduced NELSR, to our knowledge, the first benchmark for Nepali case-to-statute retrieval, consisting of 5,731 citation-derived queries from Supreme Court judgments and a corpus of 302 federal statutes. Through citation-aware query construction and legal structure-aware segmentation,

NELSR enables realistic evaluation of legal retrieval systems in Nepali. Our experiments across sparse, dense, late-interaction, and hybrid retrieval approaches show that BM25 remains a strong baseline, while dense models provide complementary semantic matching capabilities. Hybrid retrieval achieves the best overall performance by effectively combining lexical and semantic signals. We also analyze computational trade-offs among retrieval frameworks, highlighting the balance between effectiveness and resource requirements. NELSR establishes a foundation for future research in Nepali legal NLP.

7 Limitations

Although NELSR provides a standardized benchmark for Nepali case-to-statute retrieval, it has several limitations. First, the benchmark focuses on retrieval effectiveness and does not evaluate downstream tasks such as legal question answering, reasoning, or decision support. Second, relevance annotations are derived from statutory citations in judicial decisions, which provide reliable supervision but may not capture all statutes that could be relevant to a case beyond explicit citations. Third, the availability of pretrained legal retrieval models for Nepali remains limited, and our experiments rely primarily on general multilingual and Nepali-specific embedding models rather than legal-domain pretrained models. Fourth, the computational efficiency experiments were conducted on the Google Colab free-tier environment; therefore, the reported indexing time, throughput, and memory measurements should be interpreted as relative comparisons among models rather than absolute performance estimates across different hardware configurations.

8 Ethical Considerations

The data used in this research were collected from publicly accessible legal information sources in Nepal, including the Ministry of Law, Justice and Parliamentary Affairs ([Ministry of Law, Justice and Parliamentary Affairs, 2026](#)) and the Supreme Court of Nepal’s published judicial decisions through Nepal Kanoon Patrika ([Nepal Kanoon Patrika, 2026](#)). These sources provide publicly available legal documents, including statutes and judicial decisions, and do not involve private or personally collected user data. The dataset was constructed solely for academic research purposes

to advance Nepali legal information retrieval. Due to potential redistribution restrictions on the original source materials, we do not release the original legal documents or judicial decisions. Instead, we release only the extracted queries and constructed document chunks required for research evaluation, while preserving the provenance of the original sources and acknowledging the respective data providers. The released dataset is intended exclusively as a research benchmark and should not be used as a replacement for professional legal advice.

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